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FoodEx2 maintenance 2024

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Abstract

FoodEx2, developed and maintained by EFSA, is a comprehensive system for classifying and describing food and feed. It offers a standardized terminology framework to support the entire food safety and risk assessment process, from data collection across various food and feed safety domains (e.g., monitoring of pesticides residues or biological hazards) to exposure assessment. To ensure its ongoing relevance and compliance with evolving scientific and legislative requirements, regular maintenance of this system is crucial. Seven major updates were completed so far, with all performed changes documented in annual maintenance reports. This technical report outlines the results of the eighth maintenance process conducted in 2024. Updates included addition of new terms, adjustments to the hierarchical structure, introduction of a new facet category, changes to the reportability of certain terms, deprecation of term and amendments to existing terms - enriching implicit facets and reorganising the hierarchical tree structure and parent-child relationships.

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Summary

FoodEx2 is a standardised food classification and description system developed by EFSA. It aims to enhance the comparability and exchange of data across different food safety domains under EFSA's scope. To fulfil this purpose, the system requires ongoing active maintenance to ensure it remains up to date and adaptable to different domains data collection needs. Since April 2015, following the major revision (FoodEx2 revision 2), an annual maintenance of the system has been conducted.

A maintenance technical group, consisting of EFSA staff with expertise in fields such as food consumption, chemical contaminants, pesticides residues and other, was established to assess proposals and requests from users and stakeholders and determine their implementation. The first maintenance occurred in 2015 followed by seven major releases performed between 2016-2024.

During the last (eighth) maintenance process performed in 2024, the following major changes have been implemented: 1094 new terms have been added to different hierarchies and/or facets to accommodate the needs for a new data collection (notably, the One Health Surveillance), other existing data collections (such as the Avian Influenza monitoring programmes and the Chemical monitoring data collection), requests from stakeholders or requests from internal users. Eighteen terms (n= 18) have been added based on requests from the Food and Agricultural Organisation (FAO) / World Health Organization (WHO) for mapping the food consumption data within the Global Individual Food consumption data Tool – GIFT.

Amendments to different sections of the catalogue were implemented during the maintenance of 2024. For the purposes of the Food Additives and Food Flavourings Monitoring data collection, the update of the implicit F33 Legislative-classes facet continued in the 2024 maintenance window to accommodate the mapping between the FoodEx2 base terms with the legislative classes defined in the Regulation (EC) No 1333/2008 on food additives. In addition to this, a new facet category was introduced within the F33 Legislative-classes facet to meet the data reporting requirements under the Veterinary Medicinal Product Residues (VMPR) monitoring data collection and the Veterinary Medicinal Product Residues National Control Plans (VMPR NCP) data collection according to Regulation (EU) 2022/1644 and Regulation (EU) 2022/1646.

A major revision was undertaken in the 'Veterinary Drug Residues (VetDrugRes)' hierarchy in MTX (FoodEx2) to accommodate the need of data reporting under the VMPR and the VMPR NCP data collections. In particular, the hierarchy now includes three categories at top-level group (L1) which correspond to sections for food, feed and non-food matrices. The hierarchy resembles the hierarchical structure of 'Reporting' hierarchy, focusing on selected animal-based food items and non-food matrices. The parent-child relationship of some terms, which were already reportable in the VetDrugRes hierarchy, was updated to accommodate the inclusion of upper-level terms.

An extensive update of the 'Bird (as animal)' section within the F01 – Source facet was also performed in 2024. The facet was restructured and updated to meet the demands of data reporting under the Avian Influenza surveillance programmes in poultry and wild birds. The reportability of some terms was updated, as one term (n=1) is deprecated and three terms (n=3) are dismissed from different hierarchies as found redundant, duplicated or not in line with the data used.

The present report describes the outcome of the eighth maintenance process accomplished in 2024, to keep track of the changes in the system within three minor releases published in 2024 (MTX 15.1, MTX 15.2, and MTX 15.3) and the major release published in January 2025 (MTX 16.0).

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1 Introduction

2 Background and terms of reference as provided by the requestor

In 2011, EFSA released a comprehensive food classification and description system named FoodEx2 (revision 1) (EFSA, 2011a, b). This system was developed with the aim of harmonising the description of the food matrices within data collections in different food safety domains. Following the publication, a testing phase was carried out to highlight strengths and weaknesses of the classification and to identify possible issues and needs for refinement. Based on the outcome of the testing phase, EFSA revised the system and in 2015 the second revision of the food classification and description system (FoodEx2 revision 2) was published (EFSA, 2015).

According to the guidance documents on FoodEx2 (EFSA, 2011a, 2015), internal and external users should provide to EFSA proposals for implementation and maintenance to refine the system and an annual EFSA Maintenance Technical Group (TG) should be set up to evaluate the suggestions and decide on their implementation. In fact, innovation in the food market and gained experience by using the system led to the need of keeping the catalogue updated. Following the revision 2 of the system, the FoodEx2 catalogue was revised seven times based on EFSA needs and evaluation of the data providers and stakeholders' suggestions. A first maintenance took place in 2015 (EFSA et al., 2016) followed by six major releases performed between 2016-2024. The updated catalogues (MTX 10.0, MTX 11.0, MTX 12.0, MTX 13.0, MTX 14.0 and MTX 15.0) and the technical reports that summarize all changes performed during the maintenance from 2016 - 2023 were published (EFSA, 2019b, EFSA, 2020, EFSA, 2021b, EFSA, 2022, EFSA, 2023, EFSA, 2024).

Moreover, harmonized data collection in the areas of pesticides, contaminant occurrence, veterinary medicinal product residues and food additives (EFSA, 2025), as well as on zoonoses, zoonotic agents, antimicrobial resistance and foodborne outbreaks (EFSA, 2024), is reflected in the maintenance process of the FoodEx2 catalogue. Considering that all the above data domains and several other ad-hoc data collections in EFSA use the FoodEx2 catalogue (e.g., food consumption, botanicals etc.), there was a need to accommodate different requests. Some changes were performed to enable giving a clear guidance (in terms of FoodEx2 codes) to data providers before the data collection launch date, while others were requested from EFSA staff working in different domains, data providers and network members (hereafter stated as users).

By using the system, both internal and external users and stakeholders provided constructive comments and requests for addition of missing terms and other type of updates in different sections of the catalogue. All these suggestions were considered for the maintenance purposes. A technical group composed of EFSA staff with expertise in different domains carried out the maintenance process, evaluating whether and how to implement the received suggestions in the FoodEx2 system.

The present report describes the outcome of the eighth maintenance process undertaken in 2024 to keep track of the changes in the system implemented within three minor releases published in 2024 (MTX 15.1, MTX 15.2 and MTX 15.3) and the major release

published in January 2025 (MTX 16.0). To perform a correct coding, with a reasonable level of detail, this report should be read in conjunction with the technical report 'The food classification and description system FoodEx2 (revision 2)' (EFSA, 2015) where the logic and the structure of the system and the rules for the correct use of FoodEx2 are described.

3 Data and methodologies

3.1 Data

The FoodEx2 catalogue MTX 15.0 - version published in January 2024 was used as a starting point. The suggestions provided during the period between January 2024 and January 2025 by users and stakeholders of the FoodEx2 system were collected through written feedback and used to update the system.

3.2 Methodologies

Each request for additions or structural changes was carefully evaluated. Based on expert judgment, the TG determined whether a change was needed, or the request was already addressed by the existing terms and facet descriptors. All changes adhered to the system's established structure and underlying logic, as described in the report on the development of FoodEx2 revision 2 (EFSA, 2015).

3.2.1 Maintenance Process

To carry out the eighth maintenance of the FoodEx2 catalogue, the TG performed the following steps:

- Evaluation of the requests, comments and proposals from internal and external stakeholders
- Decision on the implementation of the proposals based on the following options: addition of new terms, extension or better specification of the scope of existing terms or identification of the correct way of implementing the suggestion using already existing terms and rules, change of the parent-child relationship of the terms, deprecation and setting of terms as not reportable/dismissed in one hierarchy but still reportable in others.
- Checking for inconsistencies and ambiguities
- Update of the FoodEx2 catalogue
- Summary of the changes for the present technical report.

3.2.2 Tools

The tools used for implementing the maintenance within the three minor releases published in 2024 (MTX 15.1, MTX 15.2 and MTX 15.3) were Microsoft Excel® and the FoodEx2 Catalogue Browser (EFSA, 2019a), a user-friendly tool developed in Java by EFSA staff (Integrated Data former Evidence Management Unit) to help create FoodEx2 codes and to facilitate the management of the MTX catalogue. In addition, for implementing the changes during the major release published in January 2025 (MTX 16.0), the new internal catalogue management tool developed under EFSA's Rebuild Data Framework (DF) project, was used.

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The scope notes, implicit attributes and facets of some terms were updated using the SAS (SAS Institute Inc., Cary, NC, USA) Enterprise guide 8.3.

4 Maintenance Output

The following sections provide an in-depth explanation of the maintenance operations and reasoning behind the implemented changes. This overview is mainly structured along the impacted food safety domains, stakeholders, and legislative developments driving these modifications. Additionally, horizontal changes – those not tied to specific data collection – were also performed for maintenance purposes. These impacted various aspects (e.g., term extended names, scope notes and/or implicit facets) of existing terms from different hierarchies and facets (see Appendix A). For instance, changes on the implicit facets of FoodEx2 terms related to olive oils were performed to allow the correct mapping of the FoodEx2 terms to the categories according to the Regulation (EU) 2022/2104¹ as regards the marketing standards for olive oil. Furthermore, the deprecation of a term motivated changes which were associated with the update of term extended name and/or implicit facets of existing FoodEx2 terms.

A comprehensive table detailing all changes performed in 2024, along with the affected hierarchies and facets and the rationale behind them, is available in Appendix A.

4.1 Updates for the Avian Influenza surveillance programmes and data collection

FoodEx2 is used for reporting data under the Avian Influenza surveillance programmes in poultry and wild birds, carried out by Member States, according to Regulation (EU) 2016/429 and Commission Delegated Regulation (EU) 2020/689². To meet the data reporting requirements in this domain, an extensive update of the Bird section, 'A057X – Bird (as animal)', was performed in the maintenance process in 2024 due to the addition of new birds requested by data providers to be added to the MTX (FoodEx2). This section required additional structuring in terms of completion of taxonomy and parent-child relationships to maintain the consistency of the existing hierarchical structure in FoodEx2 bird section.

An external expert in ornithology was engaged under EFSA's Individual Scientific Advisors (ISA) scheme to provide scientific and technical support in ecosystem analysis. The task involved the assessment of the new birds to be added in the FoodEx2 catalogue requested by data providers and an overall curation of the current taxonomy as well as the collection of detailed information relevant to term names, common names, scientific names, Euring codes etc. Thus, necessary changes and amendments to the current taxonomy were proposed.

To accommodate data providers' requests for the Avian Influenza data collection, n = 800 new bird species have been added and made reportable in the F01 – Source facet during the maintenance process of 2024. The newly added taxonomic entities included one order,

¹ Commission Delegated Regulation (EU) 2022/2104 of 29 July 2022 supplementing Regulation (EU) No 1308/2013 of the European Parliament and of the Council as regards marketing standards for olive oil, and repealing Commission Regulation (EEC) No 2568/91 and Commission Implementing Regulation (EU) No 29/2012

² Commission Delegated Regulation (EU) 2020/689 of 17 December 2019 supplementing Regulation (EU) 2016/429 of the European Parliament and of the Council as regards rules for surveillance, eradication programmes, and disease-free status for certain listed and emerging diseases (Text with EEA relevance). OJ L 174, 3.6.2020, p. 211–340.

six families, 25 genera, 766 species and two subspecies. Euring³ codes were attributed to 599 terms, reported as necessary for the Avian Influenza Data Collection.

This systematic revision also motivated changes in existing terms, which were associated with: 1) update or addition of new common and/or scientific names, 2) update of the term extended name, 3) removal of existing common names and/or scientific names to ensure more accurate description of the term names, and 4) alterations to parent-child relationships. Term extended names were updated for 52 codes, common names were added or updated for 134 terms, while scientific name/synonyms were added or updated for 68 terms. A new parent term was assigned to 14 codes mostly due to the addition of missing genera in taxonomy (see section 4.7.3). Consequently, scope notes were updated for 193 terms, mostly to reflect changes in common and/or scientific names (see Appendix A).

4.2 Updates for the One Health surveillance programme and data collection

FoodEx2 is used for reporting data under the One Health Surveillance data collection according to the Commission Implementing Decision of 25.7.2023 amending Implementing Decision C(2022) 317 final of 14 January 2022 on the financing of the Programme for the Union's action in the field of health ('EU4Health Programme') and the adoption of the work programme for 2022⁴. The European Commission (EC) has allocated specific resources for Member States for setting up a coordinated surveillance system under the One Health approach for cross-border pathogens that threaten the Union. The results of this surveillance should be collected by EFSA to perform a risk assessment aiming at identifying One Health zoonotic risks for the EU for which surveillance is needed and allowing for an iterative approach which would allow to review the surveillance priorities. The data for the One Health surveillance are referring to pathogen detection in livestock, wildlife, vectors and environmental samples.

To meet the data reporting requirements in this domain, during the maintenance window of 2024, extensive amendments to different hierarchies and/or facets of the MTX (FoodEx2) were implemented. These amendments were associated with the addition of new animals as natural sources. A total of 268 terms was incorporated in F01 - Source facet. In particular, 242 new terms were added to Insects section, 'A04TF - Insects (as animal)', under the category 'Diptera (as animal)' which contains the insects belonging to the taxonomic order Diptera. The newly added taxonomic entities included one family, four genera, 233 species and four subspecies. Moreover, 26 terms were added to the category 'Mites and Ticks (as animal)', which contains the animals of the taxonomic subclass Acari, and in particular the orders Ixodida (ticks) and Mesostigmata (mites). The taxonomic entities introduced under this category include two genus and 24 species. Additionally, VectorNetCode, was assigned as implicit attribute to 261 terms in the F01 - Source facet where a direct mapping was found. The 'VectorNetCode' allows a mapping between the

³ Euring is the coordinating organisation for European bird ringing schemes which maintains a databank containing ringing recovery information gathered throughout Europe.

⁴ COMMISSION IMPLEMENTING DECISION of 25.7.2023 amending Implementing Decision C(2022) 317 final of 14 January 2022 on the financing of the Programme for the Union's action in the field of health ('EU4Health Programme') and the adoption of the work programme for 2022. Available here: https://health.ec.europa.eu/document/download/18379bcf-ef74-4c7d-9614-dacf3e1d10b9_en?filename=c_2023_5052_1_act_en.pdf

new vector species introduced under the sections of 'Insects' and 'Mites and Ticks', with the VectorNetCode for vectors collected under 'VectorNet' project. VectorNet is a project of the European Centre for Disease Prevention and Control (ECDC) and EFSA which aims to contribute to improving preparedness and response for vector-borne diseases, following a 'One-Health' approach.

4.3 Updates for the Veterinary Medicinal Product Residues (VMPR) monitoring data collection in the context of Chemical Monitoring data collection

In the frame of the Chemical Monitoring data collection, EFSA receives monitoring data on veterinary medicinal product residues (VMPR) levels in food and feed. FoodEx2 is used for reporting data under the Veterinary Medicinal Product Residues (VMPR) monitoring data collection and the Veterinary Medicinal Product Residues National Control Plans (VMPR NCP) data collection according to Regulation (EU) 2022/1644⁵ and Regulation (EU) 2022/1646⁶ on the official controls of VMPR in live animals and animal products.

To meet the data reporting requirements in this domain, during the maintenance window of 2024, new legislative classes were created within the F33 Legislative-classes facet hierarchy in the MTX (FoodEx2). To allow an easy mapping of the samples to the main VMPR legislative commodity groups, 16 new categories were introduced under the new hierarchy term 'A1ANN Veterinary Medicinal Product Residues classes (Annex III – 2022/1646)'. Further details on the newly added facet categories can be found in section 4.7.8.

In addition to this, to facilitate the creation of the 'Business Rules (BR)' and thus improve the quality of data EFSA receives under the VMPR and the VMPR NCP data collections, an extensive revision of the 'Veterinary Drug Residues (VetDrugRes)' hierarchy in MTX (FoodEx2) was performed in the 2024 maintenance process. The hierarchy now includes three categories at top-level group (L1), which corresponds to sections for food, feed and non-food matrices. The feed section was introduced in the VetDrugRes hierarchy, whereas selected animal-based food items and other non-food matrices, that were reportable in different hierarchies, were also added (see further details in section 4.7.1).

Furthermore, three new terms were created and made reportable in different facet hierarchies. One term related to a fat tissue was introduced in the F02 Part-nature facet and two terms with respect to animal feed were added under the F23 Target-consumer facet (see sections 4.7.4 and 4.7.6, respectively). The addition of a new term within F02

⁵ Commission Delegated Regulation No (EU) 2022/1644 of 7 July 2022 supplementing Regulation (EU) 2017/625 of the European Parliament and of the Council with specific requirements for the performance of official controls on the use of pharmacologically active substances authorised as veterinary medicinal products or as feed additives and of prohibited or unauthorised pharmacologically active substances and residues thereof. OJ L 248, 26.9.2022, p. 3–17.

⁶ Commission Implementing Regulation (EU) No 2022/1646 of 23 September 2022 on uniform practical arrangements for the performance of official controls as regards the use of pharmacologically active substances authorised as veterinary medicinal products or as feed additives and of prohibited or unauthorised pharmacologically active substances and residues thereof, on the specific content of multi-annual national control plans and specific arrangements for their preparation. OJ L 248, 26.9.2022, p. 32–45.

Part- nature facet also led to modification in terms of common names and scope notes to one term reportable in this facet hierarchy (see Appendix A).

4.4 Updates for the new Food Additives and Food Flavourings Monitoring data collections

In accordance with Regulation (EC) No 1333/2008⁷ and Regulation (EC) No 1334/2008⁸ the consumption and use of food additives and food flavourings shall be monitored, and data shall be gathered to assess that their use does not raise a health concern. Following a new EC Mandate⁹, EFSA must collect data from the Member States, starting with a pilot in 2025, for monitoring the occurrence (analytical results via the Chemical Monitoring data collection), usage levels and presence (new data collection) of food additives and flavourings.

For this purpose, the update of the implicit facets of FoodEx2 base terms was partially implemented during the 2024 maintenance window and will continue during 2025. Specifically, a mapping between FoodEx2 base terms, reportable in different hierarchies including 'Reporting' and 'Exposure' hierarchies, with the F33- Legislative - classes defined in the Regulation (EC) No 1333/2008 on food additives was applied in the MTX (FoodEx2). The use of the regulatory facet descriptor easily allows domain-specific information to be provided, while still defining the food group in the standard way for the purpose of information storage and retrieval from databases.

The F33 – Legislative- classes facet descriptors under 'A163R Food additive classes (1333/2008)' are in accordance with the food legislative categories that are described in Part E of Annex II of Regulation (EC) No 1333/2008 on food additives. To accommodate the needs under the food additives and food flavourings data collections (analytical data, use levels and no presence data at sample level), the update of the term extended names of four food additive classes within F33 Legislative-classes facet was performed during 2024 maintenance process (see section 4.7.8 for further details) based on the amendments done in the Regulation.

4.5 Updates for the Pesticides residues data collection in the context of Chemical Monitoring data collection

The catalogue was also revised for the purposes of the Chemical Monitoring data collection, and in line with Annex I of the Regulation (EC) No 396/2005 on maximum residue levels of pesticides¹⁰. An update of the implicit attribute "matrixCode", which associates a FoodEx2 term with its corresponding legislative category specifying its applicable Maximum

⁷ Commission Regulation (EC) No 1333/2008 of the European Parliament and of the Council of 16 December 2008 on food additives. OJ L 354, 31.12.2008, p. 16–33.

⁸ Regulation (EC) No 1334/2008 of the European Parliament and of the Council of 16 December 2008 on flavourings and certain food ingredients with flavouring properties for use in and on foods and amending Council Regulation (EEC) No 1601/91, Regulations (EC) No 2232/96 and (EC) No 110/2008 and Directive 2000/13/EC. OJ L 354, 31.12.2008, p. 34–50.

⁹ Request for scientific and technical assistance in the collection and reporting of the data obtained by the Member States in accordance with Article 27 of Regulation (EC) No 1333/2008 and Article 20 of Regulation (EC) No 1334/2008. Link to the mandate: link here: <https://open.efsa.europa.eu/questions/EFSA-Q-2023-00028>

¹⁰ Official Journal of the European Union, Commission Regulation (EU) 2018/62, replacing Annex I to Regulation (EC) No 396/2005 of the European Parliament and of the Council https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:32018R0062&from=EN#ntr12-L_2018018EN.01000302-E0012

Residue Limit (MRL) was performed on 21 terms. During the maintenance window of 2024, the matrix code for three terms have been updated and for 18 terms a new matrix code was added. Table 1 summarizes the updates of the matrix codes as part of the maintenance process of 2024. Furthermore, according to the Regulation (EU) 2019/1793¹¹ on official controls of certain goods from third countries and its reviewed annexes, a new term was created ('A19RM Mukunuwenna') and placed under 'A00MY Other spinaches and similar leaves' in different hierarchies. The newly created code was assigned with the matrix code 'P0252990'.

Table 1: Updates in the matrix codes of existing FoodEx2 terms in line with the with Annex I of the Regulation (EC) No 396/2005 related to pesticides

FoodEx2 code	Term Extended name	matrixCode
A18FY	Ginger roots spice	P0213040-006
A0CGY	Horseradish roots spice and similar-	P0213040
A00QL	Horseradish roots spice	P0213040-000
A032P	Brown cane sugar	P1300000
A033T	Citrus molasses	P1300000
A033V	Sugar beet molasses	P1300000
A037X	Beef tallow (including processed suet)	P1300000
A037Z	Buffalo tallow	P1300000
A038A	Camel tallow	P1300000
A038B	Sheep tallow	P1300000
A038C	Goat tallow	P1300000
A038D	Horse tallow	P1300000
A038E	Poultry fats, processed	P1300000
A038F	Chicken fat, processed	P1300000
A038G	Duck fat, processed	P1300000
A038H	Goose fat, processed	P1300000
A038J	Turkey fat, processed	P1300000
A038S	Buffalo milk fat	P1300000
A038T	Camel milk fat	P1300000
A038Z	Goat milk fat	P1300000
A039A	Sheep milk fat	P1300000

¹¹Commission Implementing Regulation (EU) 2019/1793 of 22 October 2019 on the temporary increase of official controls and emergency measures governing the entry into the Union of certain goods from certain third countries implementing Regulations (EU) 2017/625 and (EC) No 178/2002 of the European Parliament and of the Council and repealing Commission Regulations (EC) No 669/2009, (EU) No 884/2014, (EU) 2015/175, (EU) 2017/186 and (EU) 2018/1660. OJ L 277, 29.10.2019, p. 89–129.



4.6 Terms requested by FAO/WHO

The Food and Agricultural Organisation (FAO) / World Health Organization (WHO) requested 18 terms for mapping the food consumption data within the Global Individual Food consumption data Tool (GIFT)¹². Most of these changes affect the inclusion of novel plant ('as plant') and animal ('as animal') species, in the F01 – Source facet (see Appendix A). Moreover, two existing terms, present only in the Botanicals hierarchy, were added (made reportable) in the F01-Source facet, as captured in Table 2 - Update of the F01 source facet hierarchy during the maintenance in 2024. In addition to this, coding solutions to the users were provided to better specify the data according to FoodEx2 system.

4.7 Amendments in hierarchies and facets

4.7.1 Updates of the Veterinary Drug Residues (VetDrugRes) hierarchy

The VetDrugRes hierarchy is a hierarchy that encompasses the groups used for recording data on residues of veterinary medicinal products. In this domain, the data refer to a specific animal but may range between food samples and non-food animal-related samples. In the maintenance window 2024, a revision of the top-level elements and the different food groups included in this hierarchy was performed to accommodate a request provided under the VMPP and the VMPP NCP data collections. The aim was to create a copy of the hierarchical structure of 'Reporting' hierarchy including only a subset of base terms that are of animal origin, feed and some non-food matrices. This would facilitate the creation of 'Business Rules' to meet the data reporting requirements in this domain and thus improve the quality of data EFSA receives.

Hence, the hierarchy includes three categories at top-level group (L1), which correspond to the sections for food, feed and non-food matrices. Upper - level terms and processed terms of animal origin were added under the food section. The feed section was also introduced and made reportable in the updated VetDrugRes hierarchy. Furthermore, the term related to the freshwater for animal farming was added under non-food matrices to cover the data reporting of environmental samples. The VetDrugRes hierarchy now consists of 1976 terms, including 25 hierarchy terms. This systematic revision also motivated modification in existing terms, mainly referring to the food items and non-food animal related matrices, which were associated with alterations to parent-child relationships, since already existing terms in VetDrugRes hierarchy were placed under the newly created top-level categories. In addition, the reportability of one hierarchy term was changed to align with the existing hierarchical structure of the 'Reporting' hierarchy (see section 4.8). The current view of the VetDrugRes hierarchy compared to the previous version of the catalogue (MTX 15.3) is presented in Figure 1.

¹² <https://www.fao.org/gift-individual-food-consumption>

15.3 MTX (FoodEx2 Matrix)

- Non-food animal-related matrices [A0C60]
- Mammals and birds meat [A0EYH]
- Animal fresh fat tissues [A01TT]
- Animal mechanically separated meat (MSM) [A0BY5]
- ▲ Animal offal and other slaughtering products [A04MQ]
- Fish (meat) [A026V]
- Fish offal [A02EH]
- Crustaceans [A02FD]
- Molluscs [A02GM]
- Seafood offal [A16FR]
- Sea-squirts and other tunicates [A02GN]
- Sea urchins and other echinoderms [A02GP]
- Jellyfishes and similar [A02GY]
- ▲ Terrestrial animals other than mammals and birds [A0EZT]
- Milk [A02LT]
- Whole eggs [A031F]
- ▲ Liquid egg products [A031Q]
- Honey [A033J]

16.0 MTX (FoodEx2 Matrix)

- ▲ Food [A0B6Z][NOT REPORTABLE]
 - ▲ Mammals and birds meat and products thereof [A0E5S][NOT REPORTABLE]
 - ▲ Fish meat and products thereof [A0EZR][NOT REPORTABLE]
 - ▲ Seafood and products thereof [A0EQZ][NOT REPORTABLE]
 - ▲ Terrestrial animals other than mammals and birds [A0E7T][NOT REPORTABLE]
 - ▲ Milk and milk products (dairy) [A0BXZ][NOT REPORTABLE]
 - ▲ Eggs and egg products [A031E][NOT REPORTABLE]
 - ▲ Composite dishes [A03VA][NOT REPORTABLE]
 - ▲ Isolated purified ingredients (including mineral or synthetic) [A0BXX][NOT REPORTABLE]
- ▲ Feed [A0BB9]
 - > Cereal grains and products derived thereof (feed) [A0BBA]
 - > Oil seeds, oil fruits, and products derived thereof (feed) [A0BEB]
 - > Legume seeds and products derived thereof (feed) [A0BG2]
 - > Tubers, roots, and products derived thereof (feed) [A0BH7]
 - > Other seeds and fruits, and products derived thereof (feed) [A0BJL]
 - > Forages and roughage, and products derived thereof (feed) [A0BKT]
 - > Other plants, algae, fungi and products derived thereof (feed) [A0BLK]
 - > Milk products and products derived thereof (feed) [A0BM9]
 - > Land animal products and products derived thereof (feed) [A0BMV]
 - > Fish, other aquatic animals and products derived thereof (feed) [A0BNJ]
 - > Minerals and products derived thereof (feed) [A0BP4]
 - > Fermentation (by-)products from microorganisms the cells of which have been inactivated or killed (feed) [A0BRA]
 - > Miscellaneous (feed) [A0BRR]
 - > Compound feed (feed) [A0BT0]
- ▲ Non-food matrices [A0BYQ]
 - > Freshwater for animal farming [A0F7L]
 - Non-food animal-related matrices [A0C60]

Figure 1: Update of the VetDrugRes hierarchy

4.7.2 Updates of the PRIMo hierarchy of FoodEx2

To align with the upcoming version of the Pesticide Residue Intake Model (PRIMo revision 4) tool and to reflect the refinements made on the Raw Primary Commodity (RPC) consumption database considered in the tool, the current PRIMo hierarchy was updated by dismissing two terms (see section 4.8).

4.7.3 Changes in the F01 – Source facet

F01 – Source facet was amended with a total number of 1086 new terms to allow mapping of samples within different data collections and projects (e.g., Avian Influenza, One Health surveillance programmes and data collections). Given the extensive amendment of the Bird section, 'A057X –Bird (as animal)', during the maintenance process in 2024, an additional structuring in terms of completion of taxonomy and parent-child relationship was required. Consequently, the addition of new terms requested by data providers led to the update of the parent for existing terms (n=14) in the catalogue. For more details see section 4.1. Moreover, to accommodate the requests from FAO/ WHO for mapping the food consumption data within the GIFT, the applicability of two existing plant species, which were previously reportable only in the Botanicals hierarchy, was updated and therefore were made reportable in F01- Source facet. Table 2 summarizes the updates in terms of the new parent-child relationships and the applicability of terms in the F01- Source facet hierarchy during 2024 maintenance window.

Table 2: Update of the F01 source facet hierarchy during the maintenance in 2024

FoodEx2 code	Term Extended name	Source Parent Code Old	Source Parent Code New
A171L	Egyptian Goose (as animal)	A188A	A1AXM
A173G	Squacco Heron (as animal)	A18JM	A1AKZ
A173Y	African Sacred Ibis (as animal)	A187Y	A1B0P
A174V	Common Goldeneye (as animal)	A188A	A1AXV
A175Q	Red-crested Pochard (as animal)	A188A	A1AZB

A17MX	Black-legged Kittiwake (as animal)	A058T	A1B0M
A17MZ	Little Tern (as animal)	A058T	A1B0D
A17SQ	Fulvous Whistling Duck (as animal)	A188A	A1AYE
A17YR	Gull-billed Tern (as animal)	A058T	A1AYN
A17YT	Sandwich Tern (as animal)	A058T	A1ANM
A184J	Common Myna (as animal)	A0CDS	A1AXJ
A184R	Eurasian Siskin (as animal)	A16CV	A1ANE
A18TZ	Zenaida Dove (as animal)	A18KY	A1B0K
A19DF	Temminck's tragopan (as animal)	A188Z	A1ANG
A15GR	Pueraria montana var. chinensis (Ohwi) Sanjappa & Pradeep (as plant)		A0BXQ
A15TV	Struthium sparganophorum (L.) Kuntze (as plant)		A05MX

4.7.4 Changes in the F02 –Part-nature facet

As described in section 4.3, FoodEx2 is used for reporting data under the VMPR and the VMPR NCP data collections. To allow the mapping of samples tested in this domain, the F02- Part nature was amended with one new term. Thus, a new facet descriptor for the 'A1A8D Kidney fat (as part-nature)' under the code 'A0F1B Visceral fat (as part-nature)' was added for the purposes of these data collections, to better specify and indicate the part of the animal tested. Due to the addition of the new term, modifications in terms of common names and scope notes were performed to one term (see Appendix A).

4.7.5 Changes in the F21 – Production-method facet

During the maintenance window of 2024, three new terms were created within the F21 - Production – method facet hierarchy to accommodate the needs for the upcoming baseline survey on the prevalence of methicillin-resistant *Staphylococcus aureus* (MRSA) for pigs. The new terms were introduced under the parent code 'A0CAG Farmed in single purpose holding' in F21 – Production-method facet to describe the different types of farms where pigs are growing (see Appendix A).

4.7.6 Changes in the F23 – Target – consumer facet

To accommodate the needs of data reporting under the VMPR and the VMPR NCP data collections, two new facet descriptors were added in the F23 – Target-consumer facet. This facet is recommended for feed samples under this data collection to indicate the species for whom the feed is intended if not already implicitly included in the selected base term. According to the Regulation (EU) 2022/1646 on the official controls of VMPR in live animals and animal products, three new VMPR product categories are introduced, including insects and reptiles. For these purposes, during the 2024 maintenance process, 'A1A8B Reptiles feed' and 'A1A8C Insects feed' were added under the 'A07TX Feed for food animals' in the F23 – Target – consumer facet (see Appendix A).

4.7.7 Changes in the F31 – Animal – age-class facet

According to the Regulation (EC) No 2160/2003¹³ on the control of *Salmonella* and other specified food-borne zoonotic agents, the F31 – Animal – age class facet hierarchy was updated in terms of the addition of two new terms. Respecting the minimum sampling requirements set out for all *Salmonella* serotypes with public health significance, the newly created terms referred to two phases of production which sampling must cover. The new terms were added under the parent term 'A0C8K Age classes for birds' and correspond to the four-week-old poultry and to poultry which have covered two weeks before moving to laying phase or laying unit (see Appendix A).

4.7.8 Changes in the F33 – Legislative – classes facet

Sixteen new F33 legislative classes were introduced during 2024 maintenance process to accommodate the needs under the VMPR and VMPR NCP data collections. According to Annex III of Regulation (EU) 2022/1646, the possibility of reporting processed products from the main raw product categories for samples collected under the national risk-based control plan for third-country import (Plan 3) was introduced. To allow the mapping of the samples to the main VMPR legislative commodity groups, the new F33 Legislative classes were created under the new hierarchy term 'A1ANN Veterinary Medicinal Product Residues classes (Annex III – 2022/1646)'. Table 3 presents the new Veterinary Medicinal Product Residues (VMPR) classes added in the F33 Legislative-classes facet.

Table 3: New facet descriptors added in the F33 Legislative-classes facet in line with Annex III of Regulation (EU) 2022/1646 on the official controls of VMPR in live animals and animal products

FoodEx2 code	Term Extended name
A1ANN	Veterinary Medicinal Product Residues classes (Annex III – 2022/1646)
A1ANP	VR - Bovines
A1ANQ	VR - Sheep
A1ANR	VR - Goats
A1ANS	VR - Pigs
A1ANT	VR - Horses
A1ANV	VR - Poultry
A1ANX	VR - Aquaculture
A1ANY	VR - Milk
A1ANZ	VR - Eggs
A1APA	VR - Rabbits
A1APB	VR - Farmed Game
A1APC	VR - Wild Game
A1APD	VR - Reptiles

¹³ Regulation (EC) No 2160/2003 of the European Parliament and of the Council of 17 November 2003 on the control of salmonella and other specified food-borne zoonotic agents. OJ L 325, 12.12.2003, p. 1–15.



A1APE	VR - Insects
A1APF	VR - Honey

In addition, as described in section 4.3, the term extended names of four F33 food additive classes were updated in line with Part E of Annex II of Regulation (EC) No 1333/2008 on food additives. Table 4 presents the old and new term extended names of the four facet descriptors under 'A163R Food additive classes (1333/2008)' in the F33 Legislative-classes facet.

Table 4: Updates in the term extended names of food additive classes within F33 Legislative-classes facet in line with the with Part E of Annex II of Regulation (EC) No 1333/2008 on food additives

FoodEx2 code	Old Term Extended name	New Term Extended Name
A0C1C	FA-14.2.8 Other alcoholic drinks including mixtures of alcoholic drinks with non-alcoholic drinks and spirits with less than 15 % of alcohol	FA-14.2.8 Other alcoholic beverages including mixtures of alcoholic beverages with non-alcoholic beverages and other alcoholic beverages based on distilled alcohol with alcoholic strength by volume less than 15 %
A0C1G	FA-14.2.7 Aromatised wine-based products as defined by Regulation (EEC) No 1601/91	FA-14.2.7 Aromatised wine products as defined by Regulation (EU) No 251/2014
A0C1H	FA-14.2.6 Spirit drinks as defined in Regulation (EC) No 110/2008	FA-14.2.6 Spirit drinks as defined in Regulation (EU) 2019/787
A0C1M	FA-14.2.2 Wine and other products defined by Regulation (EEC) No 1234/2007, and alcohol-free counterparts	FA-14.2.2 Wine and other products defined in Part II of Annex VII to Regulation (EU) No 1308/2013

4.8 Deprecated and dismissed terms during the maintenance 2024

During the maintenance window of 2024, the term 'A0DCH Soyabeans for consumption (dry)' was deprecated, since being a duplicate to and thus replaced by 'A015P Soyabeans (oilseed)', and therefore, cannot be used anymore (see Table 5 – List of deprecated terms during the maintenance in 2024).

Table 5: List of deprecated terms during the maintenance in 2024

FoodEx2 code	Deprecated Term Extended name	Replaced by
A0DCH	Soyabeans for consumption in (dry)	A015P Soyabeans (oilseed)

Table 6 presents the implemented changes in reportability. Specifically, two terms have been dismissed from the PRIMo hierarchy to align the current hierarchy with the new version of the PRIMo tool (revision 4) which is currently under development. In addition, to meet the data reporting requirements under the VMPPR and the VMPPR NCP data collections, one term was set as not reportable in the VetDrugRes hierarchy to align with the current tree structure of 'Reporting' hierarchy.

Table 6: List of dismissed terms during the maintenance in 2024

FoodEx2 code	Term Extended name	Hierarchies in which the term was dismissed
A00LE	Escaroles	PRIMo hierarchy
A00MD	Dandelions	PRIMo hierarchy
A0EZT	Terrestrial animals other than mammals and birds	VetDrugRes hierarchy

5 Conclusion

The present report documents the update of FoodEx2 performed within three minor releases published in 2024 (MTX 15.1, MTX 15.2, and MTX 15.3) and the major release published in January 2025 (MTX 16.0), covering requests from internal and external users.

- In total, 1112 new terms have been added to the catalogue since the last major release 15.0.
- Throughout the 2024 maintenance window, 195 scope notes have been updated, to keep their term descriptions and references up to date with current scientific knowledge and resources as well as legislative definitions.
- Modifications were performed in existing codes from different hierarchies and/or facets which were associated with update/addition of common names (n=119) and scientific names (n= 68), update of the term extended name (n= 57), and removal of existing common names (n= 16).
- New terms (n=268) were added in different hierarchies and/or facets in accordance with the One Health Surveillance data collection.
- New facet categories (n=16) were created within F33 Legislative-classes facet to allow the mapping of the samples to the main VMPP legislative commodity groups.
- Implicit facets were updated and partially implemented to FoodEx2 terms in accordance with Food Additives and Food Flavourings Monitoring.
- An extensive update of the VetDrugRes hierarchy was performed, which concluded to the inclusion of terms and changes of already existing codes in terms of parent-child relationship and reportability.
- The Birds section is amended to accommodate Avian Influenza data collection and to complete taxonomy of existing species.
- One term is deprecated while three terms are dismissed from different hierarchies as found redundant, duplicated or not in line with the data used.
- The position of some terms in the hierarchy tree was changed since a better parent-child term relationship was found mostly due to the addition of new terms.
- The applicability of some terms was extended to further hierarchies.

The complete content of the FoodEx2 system is available in Appendix B to this document, as exported spreadsheet MTX_16.0.xlsx.

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Abbreviations

BR	Business Rules
DF	Data Framework
EC	European Commission
ECDC	European Centre for Disease Prevention and Control
EFSA	European Food Safety Authority
EPPO	European and Mediterranean Plant Protection Organization
EU	European Union
FAO	Food and Agriculture Organization
FPA	Framework Partnership Agreement
FoodEx2	Food Classification and Description System
GIFT	Global Individual Food consumption data Tool
IDATA	Integrated Data
ISA	Individual Scientific Advisors
MRL	Maximum Residue Limit
MTX	FoodEx2 Matrix hierarchy
PRIMo	Pesticides Residue Intake Model
RPC	Raw Primary Commodity
TG	Technical Group
VetDrugRes	Veterinary Drug Residues
VMPR	Veterinary Medicinal Product Residues
VMPR NCP	Veterinary Medicinal Product Residues National Control Plans
WHO	World Health Organization

Appendix A – New terms added and changes made during the 2024 maintenance

Appendix A can be found in the online version of this output (in the 'Supporting information' section)

Appendix B – FoodEx2 catalogue MTX_16.0

Appendix B can be found in the online version of this output (in the 'Supporting information' section)